

Opportunity Statement

Problem

Our current review and quality control processes are reactive, rigid, and overly dependent on simple configuration

Requests are typically reviewed after fulfillment, limiting our ability to prevent errors upstream.

This structure prevents both internal quality users and customers from defining workflows that align with their own SOPs, sites, and operational models.

The result is unnecessary rework, slower turnaround times, inconsistent compliance outcomes, and reduced trust in quality processes.

Opportunity

There is a clear opportunity to move from reactive inspection to proactive prevention.

By introducing a **Proactive Review Framework** — a configurable, workflow-driven system that enables reviews before fulfillment — we can allow users to define review logic across multiple dimensions (user, site, requester, user role (TBD), and time).

This creates a flexible foundation for quality, audit, and compliance workflows that adapt to real-world operational needs, enabling errors to be caught and corrected early while reducing cycle time and manual rework.

Goal and Impact

Goal

Build a configurable review framework that allows users to define and automate pre-fulfillment review and approval logic across workflow dimensions.

Impact

- Reduce post-fulfillment errors and rework by shifting reviews earlier in the process.
- Enable customers and internal quality teams to configure review workflows aligned with their operations.

- Improve efficiency through automation of sampling, routing, and notification processes.
- Provide leadership with visibility into review progress, performance, and outcomes.
- Establish a scalable foundation for all future quality, compliance, and audit capabilities in the platform.

Hypothesis

- **If** we enable configurable, workflow-driven reviews that can operate across multiple dimensions (user, site, requester, role, time), **then** organizations will prevent more errors before fulfillment and reduce rework.
- **If** we provide automated request sampling, dedicated review queues, and configurable notifications, **then** reviewers will spend less time on manual administration and more time performing meaningful quality checks.
- **If** we introduce real-time visibility into review progress and outcomes, **then** leadership and customers will gain confidence in quality performance, compliance, and process consistency.

Press Release (Future State)

FOR IMMEDIATE RELEASE

Introducing the Proactive Review Framework for HealthSource

HealthSource has launched a new **Proactive Review Framework**, enabling users to configure and automate reviews before fulfillment.

With flexible configuration across user, site, role, requester, and request type, teams can now create workflows that reflect their operational and compliance needs — catching errors earlier and improving turnaround times.

The framework introduces automated request sampling, dedicated review queues, and real-time dashboards that provide visibility into performance, quality trends, and compliance outcomes.

For operations and quality teams, it means faster issue detection and fewer downstream corrections.

For customers, it delivers confidence that requests meet quality standards before delivery.

This is a major step toward embedding proactive quality into every part of the workflow — improving accuracy, speed, and trust across the platform.

Pre-Mortem: How This Could Go Sideways

- **Over-configuration overwhelms users.** If setup is too complex, adoption and usability will drop.
- **Quality review becomes the bottleneck.** Without clear guardrails, review workflows could slow fulfillment instead of improving it.
- **Data integrity issues.** If operational and review data models aren't aligned, dashboards and insights will lose credibility.
- **Flexibility without structure.** Excessive customization without validation could cause conflicts and inconsistent results.
- **Low adoption after launch.** Building flexibility isn't enough, it must be intuitive and clearly valuable to both internal and external users.

Prioritized Scope

In Scope

- Configurable review engine supporting user, site, requester, role, request type, and time-based rules.
- Dedicated review queue that separates quality reviews from fulfillment workflows.
- Configurable sampling and prioritization logic to balance quality with turnaround speed.
- Integrated review interface for capturing outcomes, comments, and corrective actions.
- Dashboards and reporting tools for visibility into review activity and performance trends.
- Separation of review and operational data for analytics and compliance integration (e.g., Snowflake).

Out of Scope

- Discrete request or task assignment
- Major redesign of non-quality control user interfaces.
- Replacement of downstream certification or fulfillment logic.

Acceptance Criteria

Pre-Fulfilment Sample Reservoir Configuration

1. Requests to be sampled from the request workflow to be added to the reservoir can be configured based on any combination of the criteria below. Logic between criteria is "AND"
 - User Role
 - User
 - Site
 - Requestor
 - Request Type
2. Request reservoirs support multiple sets of sampling criteria; i.e. a reservoir can be specified to include 10 PAYD requests and 10 Continuity of Care Requests
 - Each set of sampling criteria must include a maximum count of requests
 - Each set of sampling criteria must include a maximum retention time in business days
 - Each set of sampling criteria must include a due date threshold; i.e. exclude requests due in X number of days

Illustrative Sample Reservoir Configuration

Edit Audit Pool

Configure the pool filters, release types with volumes, and retention settings.

Pool Name

Attorney & Continuity Mix

Release Types & Volumes

+ Add Type 

Attorney

%   

Count: 10 requests

Continuity of Care

%   

Count: 10 requests

Configure sampling volume for each release type. Use percentage for proportional sampling or absolute count for fixed numbers.

Additional Filters

User

Site

Sarah Johnson 

All Sites 

 Pool Retention Time 

3 days

12 hours

1 day

2 days

3 days

1 week

Custom: 72 hours

After this period, unreviewed requests will automatically proceed to fulfillment.

 Exclude Requests Due Soon 



Exclude if due within:

1 day

12 hours

1 day

2 days

3 days

Custom: 24 hours

Requests due within this timeframe will skip the audit pool to ensure timely fulfillment.

3. For a request that has been sampled, if the maximum retention time has been exceeded, then it should be removed from the sampling reservoir automatically
 - Note: In circumstances in which a particular set of sampling criteria result in an incredibly low volume of requests, it will make the sampling of such requests difficult. However, this can be mitigated by increasing the maximum retention time for the criteria.
4. A request could be included in more than one reservoir
5. STAT requests are always excluded from reservoir inclusion

Pre-Fulfilment “Audit Pending” Workflow Behavior

See [Workflow Diagram](#)

1. After a request is logged but before it moves to fulfilment the request sits in an “Audit Pending Recall” state for 30 minutes during which time an operator can return the request to the “Ready to Log” state. After this time it is evaluated against the configured sampling criteria
 - If it meets any reservoir criteria, it is evaluated for potentially being audited.
 - If the reservoir count is below its threshold, the request is transitioned into “Audit Pending” and is added to the reservoir. If the reservoir count is at its threshold and the request is randomly selected for inclusion, then it is transitioned into “Audit Pending” and is added to the reservoir
 - If a request has been selected and the reservoir is already at its threshold, the request that has been in that reservoir the longest transitions to “Ready to Fulfill”
2. Once in the reservoir, HIOS/HIOM can transition the request to “Logging” or “Ready to Fulfill”
 - The system should store and calculate the total duration a HIOS/HIOM spends reviewing a request (i.e. have the review screen open), in such a way that readily supports reporting on user (auditor and audited), duration, request types, and time stamps
 - If an HIOS/HIOM “Fails” a request, they must select at least one Corrective Action or specify Other. They may specify a comment.
 - If an HIOS/HIOM “Passes” a request, they may specify a comment.
3. If a request exceeds the maximum retention time in business days, it should automatically be removed the reservoir and move to the “Ready to Fulfill” status
 - Note: Can happen via an overnight process

Metrics and KPIs

1. Audit Task Duration
 - a. Start: Auditor opens a request in “Audit Pending”
 - b. End: Auditor “Passes” or “Fails” a request
2. Count of Audit Passed Requests

- a. HIOS/HIOM clicks “Pass” request button
 - b. Broken down by:
 - i. User Role
 - ii. User
 - iii. Site
 - iv. Requestor
 - v. Request Type
3. TBD: Defect Count
4. Count of Audit Failed Requests
 - a. Hios/HIOM clicks “Fail” request button
 - i. User Role
 - ii. User
 - iii. Site
 - iv. Requestor
 - v. Request Type
 - vi. Corrective Action

Dependencies & Constraints

- **Engineering:** Configuration logic, workflow routing, notification systems, and data modeling.
- **UX Design:** Configuration UI, dashboards, and user validation testing.
- **Product:** Coordination across internal and customer-facing review workflows.
- **Constraint:** Code freeze may shift production release to early Q1.

Risks & Mitigation

- **Configuration complexity:** Offer pre-built templates and guided setup.
- **Notification fatigue:** Provide opt-in settings and batch notifications.
- **Performance degradation:** Conduct stress and scalability testing before GA.
- **Adoption risk:** Engage early pilot customers and internal users for feedback and iteration.

Release Strategy

Alpha

Pilot sampling based on user role, site, request type, and approval functionality with internal users. Validate workflow logic, usability, and reporting metrics (task duration and outcome tracking).

Beta

Expand configurability beyond user role and request-type settings, refine existing features, introduce comment and structured corrective-action capture, enhance reporting, and support growth into additional customer segments.

General Availability

Full rollout with analytics integration, performance optimization, and leadership visibility via dashboards

Rollout Control: Managed through feature toggles by site and user group for phased enablement.

Data & Analytics Plan

- Measure reduction in post-fulfillment errors and rework.
- Track review throughput, coverage, and turnaround time.
- Monitor adoption rate across user segments.
- Evaluate performance impact and customer satisfaction improvements.
- Use review activity and audit data for continuous quality insights.

Reference Resources

Documents

 DHISC Quality Auditor – HealthSource Work Queue

[Projections for UADs Avoided in Operations based on DHISC Audit Results.xlsx](#)

Deep Dive: How Reservoir Sampling Works

Reservoir sampling allows you to maintain a fixed-size random sample from a stream of items (e.g. 20 requests to audit) without needing to know how many requests will eventually arrive.

Conceptual Workflow

1. You start with a “reservoir” based on the configured sampling logic; e.g. a list of 50 requests logged by John
2. As the first 50 requests come in, the reservoir is filled directly.

3. For each subsequent request n (numbered starting at 51): Generate a random number k between 1 and n .
 - If $k \leq \text{reservoir_size}$, replace $\text{reservoir}[k]$ with the new request.
 - Otherwise ignore the request.

This produces a uniform, unbiased sample of all requests, regardless of scale or arrival order.

Sign-Off and Acknowledgement

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